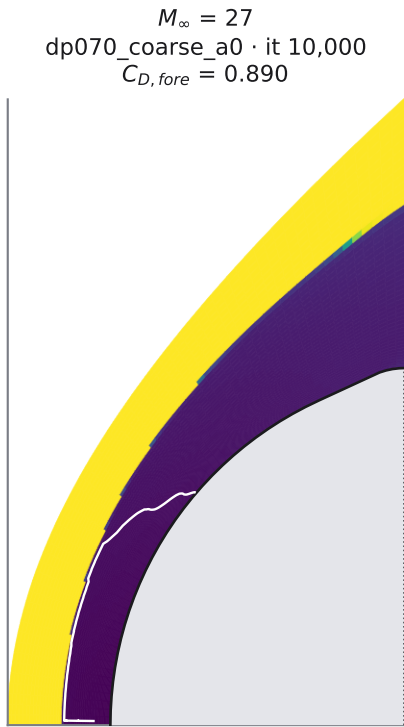
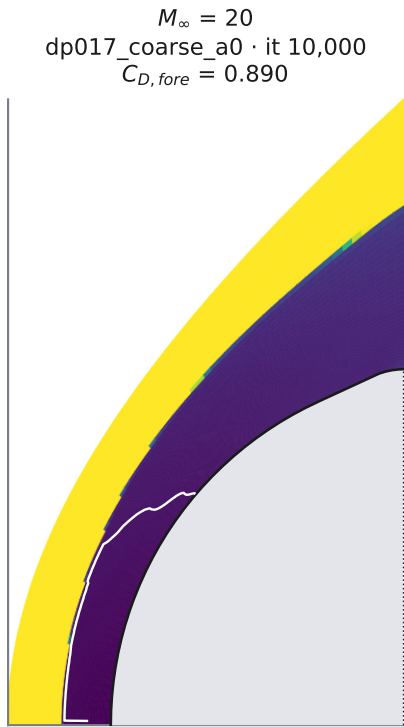


M3 design points: Mach-number field by forebody shape (rows) and Mach number (columns), coarse mesh, forebody domain; every cell's own domain, axis equal

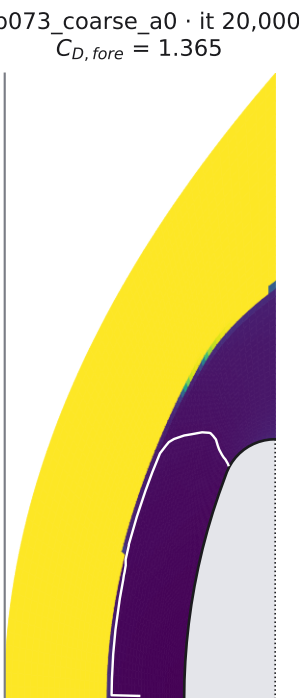
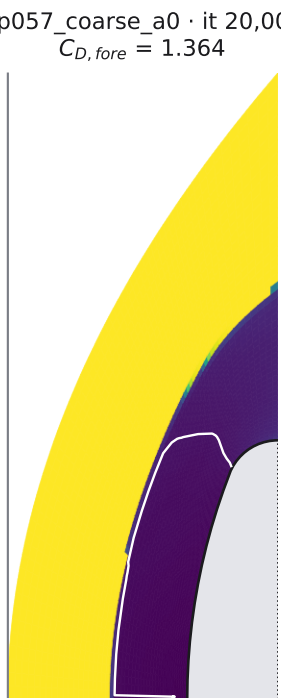
baseline
 R_n/D 0.50 · 25° · R_c/D 0.10



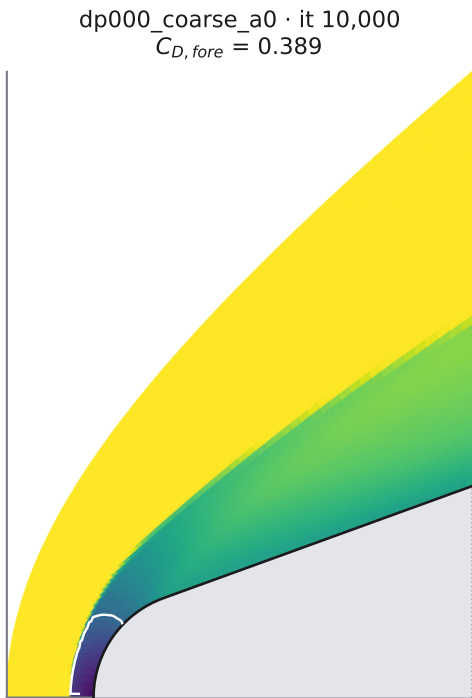
M4-front shape
(m4_front_hi)
 R_n/D 1.20 · 70° · R_c/D 0.10



no usable case
(not in the design at Mach 6)



slender sphere-cone
(vertex0)
 R_n/D 0.25 · 20° · R_c/D 0.02

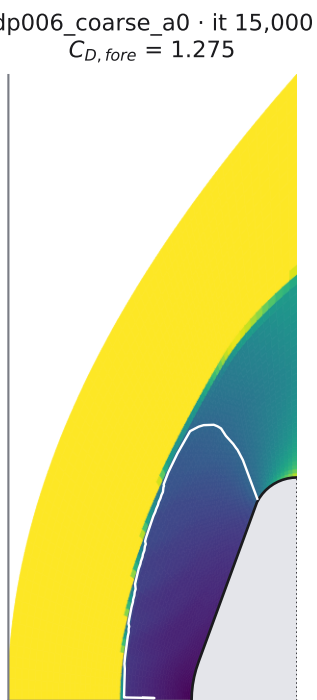


no usable case
(not in the design at Mach 6)

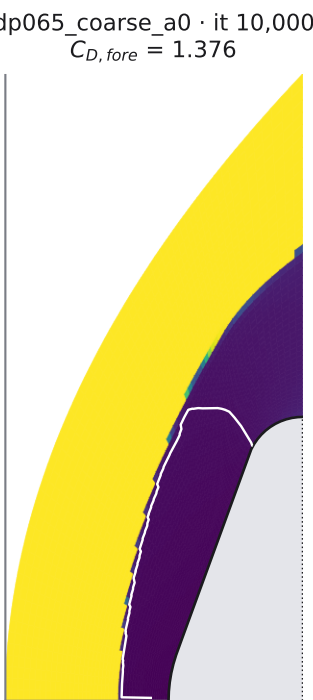
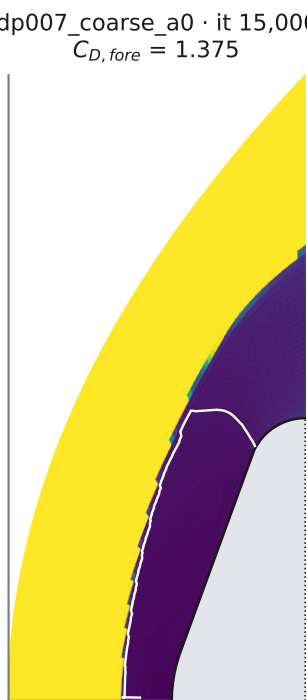


no usable case
dp062 (dp062_coarse_a2): SOLVER_FAILED
rhoCentralFoam returned -4
no field survives the van Leer hand-over (NR-28)

wide-shoulder blunt cone
(vertex3)
 R_n/D 0.25 · 70° · R_c/D 0.10

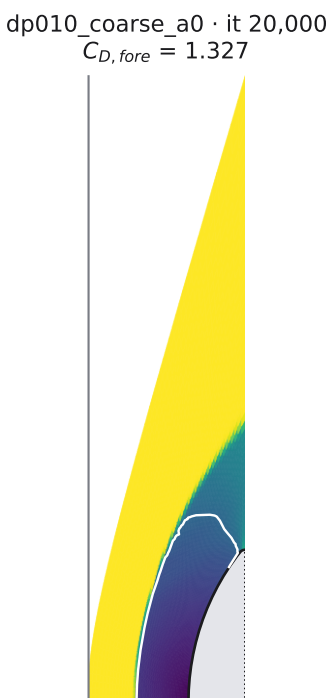


no usable case
(not in the design at Mach 6)

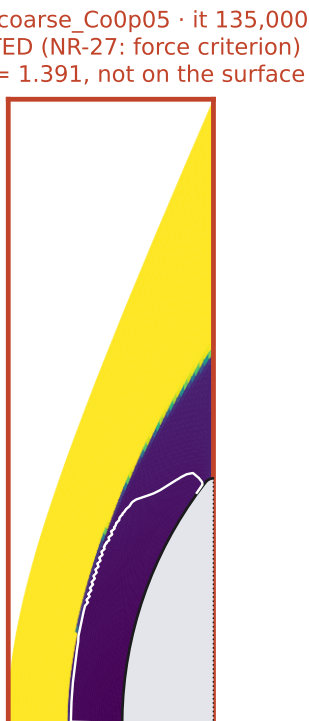


REJECTED under NR-27
(boundary0)

R_n/D 0.80 · 52.5° · R_c/D 0.02



no usable case
(not in the design at Mach 6)



dp067_coarse_Co0p05 · it 135,000
REJECTED (NR-28: force criterion)
 $C_{D,fore} = 1.395$, not on the surface



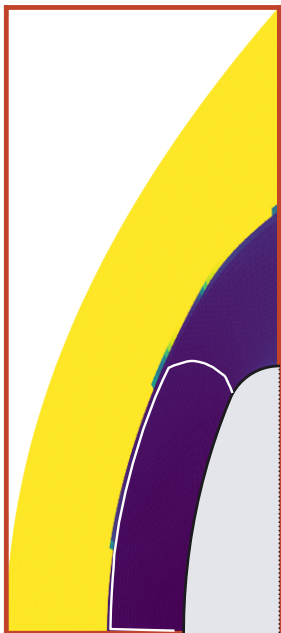
REJECTED under NR-27
(m4_front_lo_b_crit)

R_n/D 1.15 · 68.1° · R_c/D 0.10



no usable case
(not in the design at Mach 6)

dp061_coarse_Co0p05 · it 135,000
REJECTED (NR-27: force criterion)
 $C_{D,fore} = 1.370$, not on the surface



dp075_coarse_Co0p05 · it 135,000
REJECTED (NR-28: force criterion)
 $C_{D,fore} = 1.372$, not on the surface

